



BEYOND X LLC

From Mainframe to Modern Data Platforms

An Engineering Framework for Discovery, Metadata Reconstruction, Application Decoupling, Migration, Validation, AI-Assisted Knowledge Reconstruction, Cutover, and Legacy Decommissioning

Modernization is successful only when the legacy system can be retired with confidence.

Scope: IMS / CA-IDMS / VSAM / Adabas / Natural / Db2 for z/OS to relational, distributed, and cloud data platforms

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Technology with Purpose. Opportunity with Impact.
Imagination to Implementation.



Executive Summary

Mainframe modernization is often described as a database migration. That description is incomplete. A production mainframe environment is an interconnected body of data structures, application logic, batch processing, interfaces, security controls, operational procedures, reports, change history, records obligations, and institutional knowledge. Moving rows to a new database does not by itself reproduce that system.

TP-008 closes the BeyondX mainframe modernization series by treating modernization as an evidence-based engineering lifecycle: discover, reconstruct, model, decouple, migrate, synchronize, validate, cut over, stabilize, prove independence, and decommission. The framework applies both to active transactional estates and to retained-but-inactive databases maintained primarily for historical access or records retention.

Version 3 expands two themes that increasingly determine modernization success. First, a security wall is a useful control but not a substitute for supported technology, explicit access control, monitoring, and defense in depth. Second, ChatGPT and other approved AI agents can help reconstruct domain knowledge, correlate change history, identify modernization scope, generate candidate mappings and tests, and preserve institutional knowledge - while accountable humans retain approval and production authority.



Migration moves data. Modernization removes dependency.

BeyondX engineering position

- Discovery is not a phase - it is a discipline.
- Validation is not a milestone - it is continuous.
- Tools and AI accelerate modernization - they do not replace engineering judgment.
- The destination is not the cloud. The destination is freedom from the legacy dependency.

1. Modernization Landscape

Enterprises modernize from several generations of mainframe data architecture. Each source encodes structure and business meaning differently; therefore, a single schema-conversion method is insufficient.

Legacy platform	Structural characteristics	Representative modern targets
IMS	Hierarchical segments; DBD/PSB/PCB; navigational access	Db2, PostgreSQL, MySQL, cloud relational
CA-IDMS	Network model; records, sets, owners/members; subschemas	Db2, PostgreSQL, MySQL, cloud relational
VSAM	KSDS/ESDS/RRDS; copybooks; keyed/sequential access	Db2, PostgreSQL, MySQL, cloud relational
Adabas / Natural	Files, ISNs, PE/MU, DDMs, Natural logic	Db2, PostgreSQL, MySQL, cloud relational
Db2 for z/OS	Relational catalog plus z/OS SQL, utilities, batch and application dependencies	Db2 LUW, PostgreSQL, MySQL, managed cloud databases

Db2 for z/OS modernization deserves explicit treatment. Although the source is relational, migration can still require redesign or validation of SQL, stored procedures, utilities, batch workflows, encoding, security integration, operational tooling, performance patterns, and COBOL/CICS/JCL dependencies.



The migration unit is not merely a table. It is the capability, meaning, dependency, and evidence surrounding the data.

2. End-to-End Engineering Lifecycle

Stage	Engineering objective
1. Discovery	Identify data, applications, interfaces, batch, reports, security, operations, owners and dependencies.
2. Metadata Reconstruction	Reconstruct structure, business relationships, lineage, rules and hidden dependencies.
3. Target Modeling	Map semantics, keys, constraints, temporal behavior and access patterns.
4. Application Decoupling	Remove direct and indirect application dependency on the legacy platform.
5. Data Migration	Execute controlled movement with transformations and reconciliation.
6. CDC / Synchronization	Maintain change continuity where source capability and cutover strategy permit.
7. Continuous Validation	Validate structure, data, relationships, business outcomes, CDC and operations.
8. Cutover	Control final synchronization, freeze windows, routing, rollback and acceptance.
9. Stabilization	Prove performance, recoverability, observability and supportability.
10. Decommissioning	Prove independence, preserve required records/knowledge, sanitize media and retire assets.

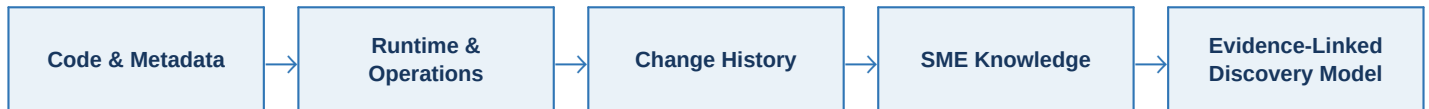


Discovery is not a phase - it is a discipline. Validation is not a milestone - it is continuous.

3. Discovery: Understand Before You Transform

Discovery must establish what the system actually does, not merely what objects exist in a catalog. Mature systems frequently encode essential relationships outside declarative database metadata.

- Inventory applications, databases, files, schemas, programs, jobs, transactions, interfaces, reports, users and owners.
- Trace upstream and downstream dependencies, including external consumers and operational feeds.
- Capture batch schedules, restart/recovery procedures, exception handling and reconciliation practices.
- Identify security boundaries, privileged access paths, service accounts, certificates, secrets and legacy protocols.
- Interview technical and business SMEs while their knowledge is still available.



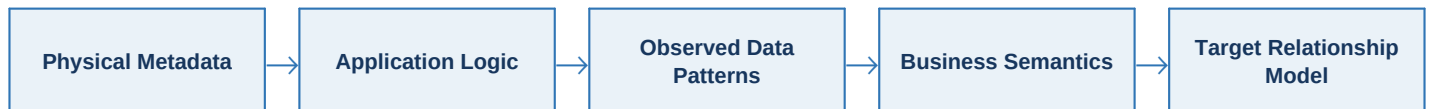
For Federal electronic information systems, records-management discovery should complement technical discovery. Required records, indexes, metadata and documentation needed to maintain and use retained information must be identified before retirement decisions are made [1].

Modernization discovery is not merely schema discovery. It is dependency reconstruction.

4. Metadata Reconstruction: Rebuild Meaning, Not Just Schemas

Source	Metadata and relationship evidence
IMS	DBDs, PSBs, PCBs, segment hierarchy, keys, SSA usage, traversal logic
IDMS	Schemas, subschemas, records, sets, owner/member relationships, CALC/keys, ADS/COBOL logic
VSAM	Catalog definitions, copybooks, keys, alternate indexes, REDEFINES, OCCURS, access patterns
Adabas/Natural	FDTs, DDMs, ISNs, PE/MU structures, Natural programs, reports and business logic
Db2 z/OS	Catalog, PK/FK, unique indexes, checks, views, triggers, procedures, packages, plans, SQL and utilities

Business relationships may be embedded in primary keys, foreign keys, unique constraints and indexes, checks, triggers, stored procedures, views, program logic, batch sequencing, reference data, naming conventions, hierarchy/set structures, access paths, and undocumented operating rules.

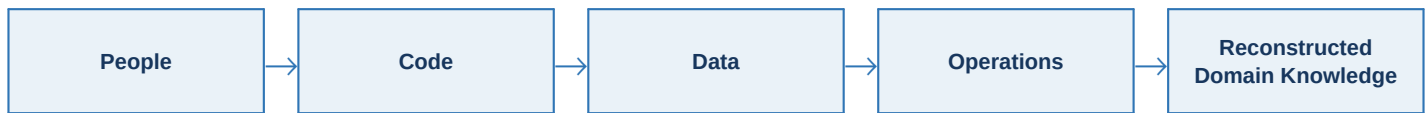


The absence of a foreign key does not mean that a business relationship does not exist.

5. Preserve Capabilities, Functionality, and Domain Knowledge

Before retirement, the modernization team should document what the legacy system provides to the mission: business capabilities, transaction behavior, calculations, reports, interfaces, batch functions, regulatory outputs, exception handling, security controls, recovery procedures, and historical-data access.

- Support risk: fewer specialists may be available during production incidents.
- Knowledge-concentration risk: critical operations may depend on a handful of individuals.
- Change risk: poorly understood systems become increasingly difficult to modify safely.
- Recovery risk: restoration may depend on undocumented procedures or obsolete tooling.
- Modernization risk: delaying transformation can reduce the knowledge available to perform it.
- Business-rule loss: logic embedded in decades of code, data structures, reports and operations may disappear.

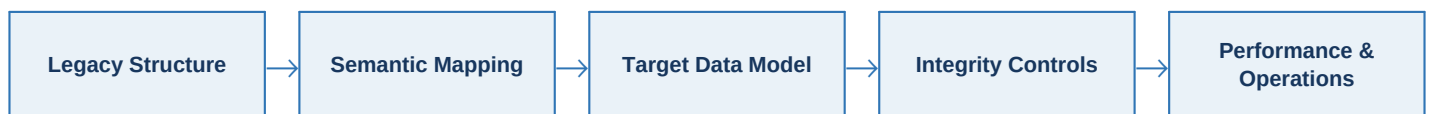


The most irreplaceable component of a legacy system may be the institutional knowledge required to understand it.

6. Target Architecture and Relational Transformation

Target design should not mechanically flatten every legacy construct. The goal is to preserve semantics while improving maintainability, integrity, performance, security and operational simplicity.

- Transform IMS hierarchy into explicit relational entities and parent/child keys while preserving ordering and optionality where meaningful.
- Translate IDMS set ownership/member semantics into relational keys, associative entities, constraints and application services.
- Map VSAM keys, alternate access paths, copybook structures and record variants into normalized or purposefully denormalized models.
- Normalize Adabas PE/MU structures and preserve stable identity mappings for ISNs where required.
- For Db2 z/OS, evaluate schema portability, SQL behavior, sequences/identity, stored logic, utilities, isolation, partitioning, encoding and performance characteristics before selecting the target.



Target architecture should preserve business meaning explicitly rather than reproducing legacy physical structures mechanically.

7. Application Decoupling

Data can be fully loaded into a new database while the enterprise remains dependent on the mainframe. COBOL programs, Natural applications, CICS transactions, batch jobs, reporting, downstream extracts, operational procedures or reconciliation processes may continue to use the legacy platform.

Application decoupling therefore requires a dependency-by-dependency retirement plan. Direct database calls, file transfers, message flows, batch extracts, terminal transactions, reporting paths and operational access must either move, be replaced, be archived, or be formally retired.



A migrated database with active legacy dependencies is not a decommissioned system.

8. Data Migration, CDC, and Cutover

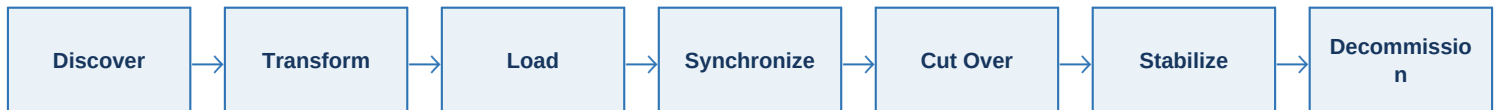
Pattern	When it fits	Key controls
Bulk extract + outage	Manageable data size and accepted downtime	Freeze, reconciliation, restart, rollback
Bulk + incremental sync	Large estates with controlled deltas	Watermarks, repeatability, delta validation
Log/journal CDC	Supported source and low-downtime objective	Ordering, latency, DDL handling, delete fidelity
Parallel operation	High assurance or staged application transition	Dual-run reconciliation and ownership rules
Final delta cutover	Most online migrations	Quiesce, final sync, validate, route, observe, rollback window



Migration patterns should reflect source capabilities, data volume, outage tolerance, business criticality and rollback requirements. Not every legacy source supports change capture in the same manner; the design should avoid pretending that all technologies provide identical CDC semantics.

9. Continuous Validation

Dimension	Examples
Structural	Objects, datatypes, keys, constraints, indexes, partitions, encoding
Data	Counts, aggregates, nulls, precision, dates/times, hashes, boundary values
Relationship	Parent/child, sets, referential integrity, PE/MU mappings, app-enforced relationships
Business	Calculations, eligibility, reports, rules, exceptions, regulatory outputs
CDC	Insert/update/delete fidelity, ordering, latency, reconciliation
Application	Old/new output comparison, transactions, interfaces, batch behavior
Operational	Backup/recovery, observability, performance, support procedures
Decommission	No remaining dependencies; records/security/knowledge obligations closed



Validation is not a milestone - it is continuous. Each stage should leave evidence that can be reused in the final decommissioning decision.

10. Out-of-Support Mainframes and Security Risk

A mainframe is not insecure merely because it is old. A supported, properly configured and actively maintained mainframe can provide strong security. The risk changes when operating systems, databases, middleware, runtimes, applications, protocols or hardware components fall out of vendor support or can no longer receive normal remediation.

Operational stability should not be confused with security support. An out-of-support component may continue running reliably while patch availability, tooling compatibility, recovery options and specialist knowledge deteriorate.

- Known or newly discovered weaknesses may remain unremediated when fixes are unavailable for the deployed release.
- Modern identity, MFA, PAM, SIEM, encryption and monitoring integrations may be harder to implement on older stacks.
- Legacy or insufficiently protected protocols can expose credentials or data if permitted internally.
- Hardware, software media, licenses, drivers and specialized recovery expertise may become difficult to obtain.
- Unsupported components can complicate risk acceptance, audit evidence and compliance obligations.



A system can be operationally stable and still represent increasing security, support, recovery, and institutional-knowledge risk.

11. Beyond the Security Wall: Defense in Depth and Zero Trust

A perimeter firewall or isolated security zone can reduce reachability, but it should not be treated as proof that an unsupported legacy platform is adequately protected. NIST Zero Trust Architecture states that network location alone does not imply trust; access to enterprise resources should be explicitly authenticated, authorized, and protected [2]. CISA guidance describes microsegmentation as a way to reduce attack surface, limit lateral movement, and improve visibility [3].

The concern is not that a modern supported mainframe is inherently insecure. The concern is indefinite reliance on perimeter isolation while patching, supportability, identity controls, protocol protection, monitoring, and institutional knowledge deteriorate.

Perimeter-only assumption	Defense-in-depth / Zero Trust transition
Firewall / security zone = trusted interior	Network location does not establish trust
Broad internal reachability	Segment and minimize permitted paths
Static credentials / broad privilege	MFA, PAM, least privilege, periodic review
Limited legacy visibility	Central logging, SIEM/NDR, anomaly review
Legacy protocol accepted because internal	Use strongest supported encryption / protected gateways
Isolation becomes permanent	Compensating controls + modernization exit plan



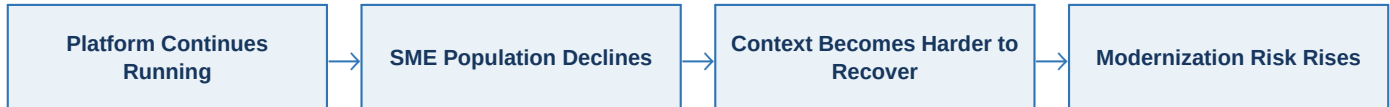
A security wall can reduce reachability. It cannot restore vendor support, produce missing security fixes, replace disappearing expertise, modernize cryptography, or eliminate dependency on an obsolete platform.

Where direct modernization is not immediately feasible, use segmentation or microsegmentation, deny-by-default connectivity, controlled administrative paths, MFA and privileged-access management where supported, least privilege, secure communications, centralized logging, configuration and vulnerability review, and tested backup/recovery. These are compensating controls and should be tied to an explicit modernization or retirement plan.

12. Workforce Scarcity and the Knowledge Clock

Organizations face a second clock: not only the age of the technology, but the availability of people who understand it. Expertise in IMS, IDMS, VSAM, Adabas/Natural, Db2 z/OS, COBOL, PL/I, Assembler, JCL, CICS, utilities and site-specific operational practices can become concentrated in a shrinking group.

The deeper risk is not simply finding someone who knows a language. It is finding someone who understands organization-specific business rules, data anomalies, interfaces, job sequences, recovery practices and design decisions accumulated over decades.



Modernization discovery should preserve knowledge while SMEs are available. Structured interviews, dependency walkthroughs, code/metadata analysis, runbooks, business-rule catalogs and validation examples should become formal modernization artifacts.

Technology may keep running while the knowledge required to understand it disappears.

13. Federal Records, Decommissioning, and Knowledge Preservation

For Federal systems, decommissioning must be coordinated with records-management, security, privacy, legal, mission and operational obligations. System retirement does not itself authorize destruction of Federal records. Required records and metadata must remain managed under applicable records schedules and disposition authorities.

- Mission/business capabilities and owners.
- Application functionality, transactions, calculations, reports and regulatory outputs.
- Data elements, definitions, classifications, retention and historical meaning.
- Primary keys, foreign keys, unique indexes/constraints, checks, triggers, views, procedures and application-enforced relationships.
- IMS hierarchy, IDMS sets, VSAM access structures, Adabas PE/MU semantics and Db2 metadata.
- Interfaces, batch schedules, upstream/downstream dependencies and external consumers.
- Security controls, identities, privileged paths, certificates, secrets, audit evidence and residual data.
- Technical and business documentation needed to understand retained records and reconstructed domain knowledge.



Do not decommission what you do not yet understand.

14. Decommissioning Readiness Matrix

Dimension	Question to close	Evidence
Capability	Has every required capability been replaced, archived or formally retired?	Capability map + owner approval
Application	Are all programs, transactions, jobs and reports accounted for?	Dependency inventory + test results
Data	Are required current/historical records complete and governed?	Reconciliation + retention mapping
Relationships	Are keys, constraints and hidden relationships preserved?	Metadata crosswalk + integrity tests
Interfaces	Are upstream/downstream consumers migrated or retired?	Interface sign-offs + traffic evidence
Records	Are schedule, hold, transfer and disposition obligations addressed?	Records-management approval
Security	Are access paths, identities, secrets, backups and residual media closed?	Security closure + sanitization evidence
Knowledge	Is sufficient domain/operational knowledge preserved?	Runbooks, models, SME validation
Operations	Can the target be supported, recovered and monitored independently?	Operational readiness evidence

Decommissioning is an evidence decision, not simply a project date.

15. Media Sanitization and Final Asset Retirement

Decommissioning includes more than turning off compute. Storage media, backups, replicas, exports, temporary files, credentials, encryption keys and removable media must be addressed according to information sensitivity and organizational requirements.

NIST SP 800-88 Rev. 2 frames media sanitization as a program for rendering access to target data infeasible for a given level of effort and selecting sanitization/disposal controls based on information sensitivity [4]. The specific method should be determined under applicable agency or organizational policy and media type.

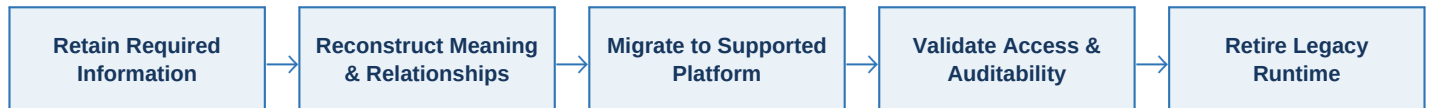


The retirement package should retain evidence of what was preserved, transferred, archived, sanitized or destroyed, by whom, under which authority, and when.

16. Retained-but-Inactive Legacy Databases

Organizations sometimes keep obsolete database platforms online after transaction processing has ended because historical information remains subject to retention, audit, legal, analytical, or business-access requirements. The workload may be passive, read-only, or rarely queried, yet the organization continues to fund the original platform.

A representative anonymized field scenario is an older Db2 for z/OS environment retained after active processing ceased. Even when isolated behind a security zone, continued operation can require mainframe capacity, storage, backup and recovery, security controls, monitoring, licensing/support arrangements, and scarce specialist skills. The retention obligation applies to information and its required context; it does not automatically require indefinite retention of the original technology.



- Distinguish active transactional, active analytical/reporting, and retained-but-inactive estates.
- Preserve provenance, keys, relationships, code/value definitions, retention schedules, legal holds, and audit evidence.
- Size the target for actual access patterns rather than reproducing legacy capacity by default.
- Select a supported Db2 LUW, PostgreSQL, MySQL, managed relational, analytical, or archival architecture according to requirements.

A data-retention requirement does not necessarily imply a platform-retention requirement.

17. AI Agents for Migration and Modernization Engineering

AI agents can operate as engineering accelerators across the modernization lifecycle. ChatGPT and other approved agents can assist with large-scale analysis of COBOL, PL/I, Natural, JCL, CICS definitions, copybooks, IMS DBD/PSB/PCB artifacts, IDMS schemas/subschemas, Adabas FDT/DDM metadata, Db2 catalogs, SQL, procedures, triggers, interface specifications and operational documentation.

Figure 1 - AI-assisted modernization engineering

AUTHORIZED KNOWLEDGE SOURCES	CHATGPT / APPROVED AI AGENTS	ENGINEERING OUTPUTS
COBOL / PL/I / Natural	Parse & explain	Dependency graph
JCL / CICS / Copybooks	Correlate dependencies	Target mappings
IMS / IDMS / VSAM / Adabas	Infer candidate mappings	Reconciliation SQL
Db2 metadata / SQL	Generate candidate SQL/tests	Test catalog
Architecture & runbooks	Summarize anomalies & gaps	Evidence-linked documentation

HUMAN ENGINEERING REVIEW & CONTROL: validate evidence, approve scope, authorize production change, and approve decommissioning.

Generated mappings, inferred relationships and proposed transformations are candidates until validated against authoritative metadata, runtime behavior, data profiling, business rules and SME knowledge.

AI proposes, correlates, explains and accelerates. Authorized humans validate, approve and control.

18. AI Agents for Application and Business Domain Knowledge

Legacy systems contain domain knowledge distributed across code, data structures, reports, tickets, procedures and people. AI agents can help reconstruct this knowledge by correlating technical artifacts with business terminology and observed behavior.

Instead of asking only what tables exist, teams can ask: Where is CUSTOMER_STATUS created, validated, modified, reported and transmitted? Which applications depend upon this IMS segment? Why does a unique index represent a business rule absent from formal documentation? Which reports or downstream feeds consume a field that appears obsolete?

Figure 2 - From fragmented artifacts to an enterprise legacy knowledge model

AUTHORIZED KNOWLEDGE SOURCES	CHATGPT / APPROVED AI AGENTS	ENGINEERING OUTPUTS
Source code	Entity & terminology extraction	Business capabilities
Database metadata	Cross-artifact correlation	Data definitions
Reports & interfaces	Business-rule hypotheses	Rules & exceptions
Tickets / procedures	Question generation	Owners & consumers
SME interviews	Evidence/provenance linking	Validation requirements

HUMAN ENGINEERING REVIEW & CONTROL: validate evidence, approve scope, authorize production change, and approve decommissioning.

Before AI transforms legacy code, it can help organizations understand what that code actually represents.

19. AI Agents and Mainframe Change-Control History

Change-control repositories often contain the missing explanation for why legacy logic exists. Source management and release systems such as Endevor, Changeman, ISPW and Git may be accompanied by change requests, incident/problem tickets, approvals, implementation notes, rollback instructions and release packages.

With appropriately authorized read-only access, an AI agent can correlate current code with historical changes to help answer why a field, validation, index, interface or exception was introduced; which production incident drove the change; whether multiple components historically changed together; and which teams possess relevant knowledge.

Figure 3 - AI-assisted change-history reconstruction

AUTHORIZED KNOWLEDGE SOURCES	CHATGPT / APPROVED AI AGENTS	ENGINEERING OUTPUTS
Endevor / Changeman / ISPW / Git	Version correlation	Why a rule exists
Change requests	Intent extraction	Historical rationale
Incident / problem tickets	Timeline reconstruction	Related components
Release packages	Co-change analysis	SME / owner leads
Approvals & rollback notes	Candidate impact reasoning	Regression-test targets

HUMAN ENGINEERING REVIEW & CONTROL: validate evidence, approve scope, authorize production change, and approve decommissioning.

Discovery access should be least-privileged, logged and read-only where practical. AI analysis should never bypass established production change-control approvals.

Change history can explain intent that source code alone cannot reveal.

20. AI Agents for Modernization and Decommissioning Scope

Modernization scope should not begin and end with a list of tables. An AI-assisted scope process can correlate business capabilities, applications, programs, transactions, databases, files, data elements, constraints, interfaces, reports, jobs, change history, records obligations, security dependencies and target capabilities.

Figure 4 - AI-assisted scope completeness

AUTHORIZED KNOWLEDGE SOURCES	CHATGPT / APPROVED AI AGENTS	ENGINEERING OUTPUTS
Capabilities & applications	Build scope graph	Target disposition
Programs / jobs / transactions	Find missing mappings	Validation evidence
Data & relationships	Surface contradictions	Unresolved gaps
Interfaces & reports	Trace consumers & owners	Named accountable owner
Records / security / change history	Generate closure questions	Decommission readiness

HUMAN ENGINEERING REVIEW & CONTROL: validate evidence, approve scope, authorize production change, and approve decommissioning.

For decommissioning, the agent can act as a completeness assistant: search the evidence base for unresolved dependencies and produce candidate questions for accountable owners. Final scope and retirement approval remain human decisions.

The objective is not to prove that migration ran; it is to prove that nothing required was left behind.

21. ChatGPT as an Authorized Modernization Engineering Partner

ChatGPT can serve as a conversational engineering layer over approved modernization knowledge sources when deployed within the customer security, privacy, records and access-control boundaries. Its role can span discovery, explanation, correlation, documentation, test design and evidence review.

A client can define the scope of what ChatGPT is permitted to access: selected source repositories, metadata catalogs, change-control history, architecture documents, tickets, test results, records schedules and sanitized SME interview transcripts. Access should follow least privilege, classification rules, logging and organizational AI governance.

Figure 5 - Authorized ChatGPT modernization architecture

AUTHORIZED KNOWLEDGE SOURCES	CHATGPT / APPROVED AI AGENTS	ENGINEERING OUTPUTS
Approved repositories	ChatGPT conversational reasoning	Discovery model
Metadata catalogs	Evidence retrieval	Scope map
Change-control history	Dependency correlation	Mappings & tests
Architecture / tickets	Candidate artifact generation	Knowledge repository
SME knowledge	Gap / contradiction analysis	Decommission evidence

HUMAN ENGINEERING REVIEW & CONTROL: validate evidence, approve scope, authorize production change, and approve decommissioning.

Important conclusions should remain traceable to authoritative artifacts. Analytical assistance should be separated from privileged execution and production change authority.

Human Engineering Review and Control surrounds the AI lifecycle.

22. AI-Assisted SME Knowledge Capture Before Retirement

Experienced mainframe engineers and business specialists often carry knowledge that is not recorded elsewhere. Structured AI-assisted interviews can help preserve this knowledge before workforce transitions remove access to it.

An SME might explain that a job normally runs after a claims extract, except at year-end when Finance provides a separate file. An AI agent can convert that narrative into candidate structured knowledge: capability, job, predecessor, input, output, data elements, exception, owner, frequency, downstream dependency and validation requirement. The SME then reviews and corrects the result.

Figure 6 - AI-assisted SME knowledge capture

AUTHORIZED KNOWLEDGE SOURCES	CHATGPT / APPROVED AI AGENTS	ENGINEERING OUTPUTS
SME interview / transcript	Structure narrative	Validated knowledge cards
Runbooks	Extract rules & exceptions	Dependency model
Code & metadata	Ask follow-up questions	Exception catalog
Reports / examples	Correlate evidence	Recovery procedures
Operational incidents	Flag uncertainty	Test scenarios

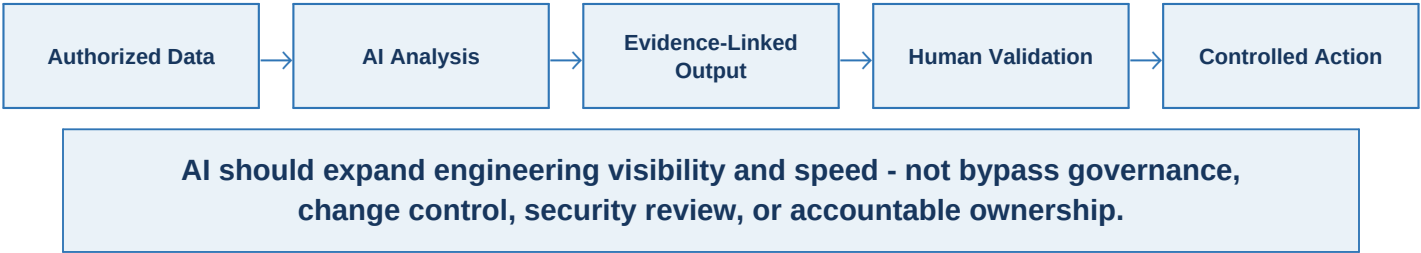
HUMAN ENGINEERING REVIEW & CONTROL: validate evidence, approve scope, authorize production change, and approve decommissioning.

AI can help preserve institutional knowledge before modernization removes the systems - or retirement removes the people - that contain it.

23. AI Governance, Security, and Human Control

AI-assisted modernization can touch highly sensitive source code, data definitions, operational procedures and security information. The AI architecture must therefore be scoped and governed as deliberately as any other engineering platform.

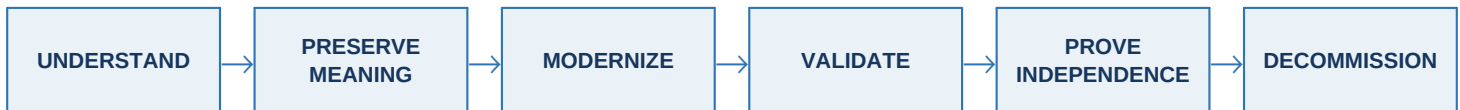
Control area	Engineering expectation
Authorization	Use approved AI services and explicitly approved repositories/data classes.
Least privilege	Grant only the information and actions required for the assigned modernization task.
Read-only discovery	Prefer read-only access for analysis; separate analysis from execution.
Provenance	Link important conclusions back to authoritative artifacts and evidence.
Human approval	Require accountable review for mappings, code, tests, scope closure and retirement.
Logging	Record access and significant agent actions according to organizational policy.
Production boundary	Do not allow autonomous production change or decommissioning authority.
Data handling	Apply customer classification, privacy, records, retention and contractual controls.



24. Capstone: Evidence-Based Decommissioning

TP-008 closes the mainframe series with a simple but demanding success criterion: the legacy system should be retired only when the organization has evidence that required capabilities, information, relationships, records, interfaces, controls and knowledge have been preserved, replaced or formally dispositioned.

This criterion applies whether the source is IMS, CA-IDMS, VSAM, Adabas/Natural, Db2 for z/OS, or a retained-but-inactive historical database. It also applies whether the target is Db2 LUW, PostgreSQL, MySQL or another supported modern platform.



- Discovery remains active until dependencies are closed.
- Validation spans structure, data, relationships, business outcomes, security and operations.
- AI accelerates evidence gathering, but accountable humans approve retirement.
- Records, metadata, provenance and domain knowledge that must survive the platform are preserved.

Modernization is successful only when the legacy system can be retired with confidence.

25. BeyondX Closing Principles

- Discovery is not a phase - it is a discipline.
- Validation is not a milestone - it is continuous.
- Metadata reconstruction must recover business meaning, not merely technical definitions.
- Application decoupling is as important as data migration.
- Security controls must protect resources during transition; network location alone is not trust.
- Isolation is a compensating control. Decommissioning removes the dependency.
- Knowledge preservation is part of architecture, not merely staffing.
- AI and automation accelerate modernization, but engineering judgment remains accountable.
- Modernization is successful only when the legacy system can be retired with confidence.

The destination is not the cloud. The destination is freedom from the legacy dependency.

References

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<https://doi.org/10.6028/NIST.SP.800-88r2>

Additional agency-specific records schedules, security policies, privacy requirements, legal holds, contractual obligations, system authorization requirements, and AI governance policies should be evaluated for each modernization program.